DSP Project Report

EOG (UP & DOWN)

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*Report on EOG Classification Steps *

Loading Data:

Function: read_file(file_path)

Details:

1- Reads a file containing EOG signal data, where each line represents a single EOG signal. Each line is split into numeric values and stored in arrays.

2- Invalid Lines: Skipped with error handling to ensure proper format.

Purpose: This function ensures that raw EOG data is properly loaded and formatted for subsequent processing steps

Output: An array of arrays, where each sub-array contains the numeric values of a single signal

Preprocessing:

1- Mean Removal: Removes the DC (Direct Current) component of the signal by subtracting the mean value of the signal from each data point

To centering the signal around zero (X-axis) makes it easier to analyze the fluctuations in the data, which are crucial for classification.

2- BandPass Filtering: A 4th-order Butterworth filter is applied with cutoff frequencies set at 0.5 Hz and 20 Hz.

Removes low-frequency drift and high-frequency noise, retaining the relevant frequency components that are characteristic of eye movements.

3- Normalization:

The signal is scaled to fit within the range [-1, 1] using min-max normalization

4- Resampling: Signals are resampled (down) to a uniform sampling frequency of 50 Hz. To ensure consistency in signal length and resolution, which is critical for feature extraction and model training and reduce computation.

Feature Extraction:

1-Wavelet Decomposition: Wavelet decomposition is performed using the Daubechies (**dp4**) wavelet up to level 4.

How the Wavelet and Levels Were Chosen:

dp4 wavelet was selected because it effectively captures the transient nature of EOG signals. Daubechies wavelets are widely used in biomedical signal processing due to their compact support and orthogonality.

The decomposition level (**max-level = 4**) was chosen based on the length of the signals and the need to capture both low and high-frequency components without over-decomposition.

https://www.mdpi.com/1424-8220/23/3/1235?utm_source=chatgpt.com

From the wavelet coefficients at each level, the following statistical features are calculated:

Mean, Standard Deviation, Maximum and Minimum: to capture the extreme values in the coefficients.

Purpose: These features summarize the frequency content and amplitude characteristics of the signal, making them suitable for distinguishing between different EOG classes

Training the Model:

Models Used:

1- K-Nearest Neighbors (KNN):

- Tuned with (**k = 3**) to balance between simplicity and accuracy.
- Good for small datasets with well classes. **(Accuracy = 80%)**

K Value	**Accuracy**
3	80%
5	73%
7	73%
9	77%
11	67%

2- Logistic Regression:

- Regularization parameter (**C= 0.1**) is used to prevent overfitting.
- Effective for binary classification tasks. **(Accuracy = 87%)**

3- Random Forest:

- Parameters included (**n-estimators = 100**) (number of trees) and (**max_depth=10**) to control overfitting.
- Robust to noise and performs well on structured data.
- **(Accuracy = 96%)**

4- (Support Victor Machine) SVM:

- **Linear** kernel is used with regularization parameter (**C=1**)
- **Probability** estimation is enabled to provide confidence scores for predictions. **(Accuracy = 93%)**

Training Steps:

- Data from **Up_Train** and **Down_Train** is preprocessed and transformed into feature vectors.
- Each model is trained using the extracted features.
- Cross-validation is employed to assess model performance and tune hyperparameters.

Evaluating the Model:

1- Test Data:

Test signals are processed using the same pipeline as the training data, ensuring consistency.

2- Prediction:

Each trained model predicts the class (Up/Down) of the test signals based on the extracted features. And give the accuracy.

Figures:

Add plots of raw and processed signals to visualize the impact of preprocessing.

Include histograms or boxplots of extracted features.